JanGyan — Learn Without Limits

Category 6: Environment & Civic Responsibility

Module 3 - Water Conservation & Management

Module Overview

This handbook is your complete, beginner-to-advanced guide to **Water Conservation & Management** — how India's water resources work, why they are under growing stress, and the practical steps every household, farmer, and community can take to conserve them. Whether you are a student, a homemaker, a farmer, or a civic volunteer, this handbook is written so you can read it once and walk away confident and capable.

By the end of this module, you will not need to search anywhere else to understand the fundamentals of water conservation in India.

Learning Objectives

After completing this module, you will be able to:

- Explain the water cycle and India's sources of freshwater
- Describe why India is classified as a water-stressed country
- Practise rainwater harvesting and groundwater recharge techniques
- Identify major causes and effects of water pollution
- Apply practical household and community water-saving habits
- Understand industrial and agricultural water management practices
- Recognise the government bodies and laws governing water in India
- Know exactly how and where to report water wastage or pollution

Why This Topic Matters

Water sustains every part of Indian life — drinking, agriculture, industry, and ecosystems — yet India holds only about 4% of the world's freshwater resources while supporting nearly 18% of the world's population. This imbalance places India among the most water-stressed large economies on Earth.

> ***Reality Check:*** *India's average annual per capita water availability stood at approximately 1,513 cubic metres in 2024 (Central Water Commission) — below the internationally recognised water-stress threshold of 1,700 cubic metres — and is projected to fall further to around 1,367 cubic metres by 2031. Groundwater extraction nationally already stands at over 60% of annual recharge, with several regions far beyond sustainable limits.*

Water conservation is not a distant policy concern — it is a daily practice that begins with a tap left running, a leaking pipe left unfixed, or a rooftop that could be harvesting rain but isn't. Every citizen's habits directly affect the water available to their community tomorrow.

This is why JanGyan continues its Environment & Civic Responsibility category with this essential, practical module.

Introduction

Water conservation is the practice of using water efficiently and protecting water resources to meet both current and future needs. **Water management** is the broader system of planning, developing, distributing, and regulating water resources — from a household tap to an entire river basin.

In simple words: water conservation is what each of us does to use less and waste less; water management is how governments, industries, and communities organise water supply so that it reaches everyone fairly and sustainably.

India's water challenge is not primarily about total quantity — the country receives substantial monsoon rainfall each year — but about **distribution, storage, and waste**. Much of the monsoon's water runs off unused into the sea because of insufficient storage and recharge infrastructure, while groundwater is extracted far faster than it is replenished in many regions.

Fundamental Concepts

Before going deeper, let us understand a few foundational ideas that the entire subject of water conservation is built upon.

The Water Cycle

The water cycle continuously recycles water between the atmosphere, land, and oceans through four main stages.

Stage	What Happens
Evaporation	Heat from the sun turns surface water from oceans, rivers, and lakes into water vapour
Condensation	Water vapour cools and forms clouds in the atmosphere
Precipitation	Water falls back to Earth as rain, snow, or hail
Collection	Water gathers in rivers, lakes, and underground aquifers, ready to begin the cycle again

Key Definitions

- **Water Stress** — When annual per capita water availability falls below 1,700 cubic metres.
- **Water Scarcity** — When annual per capita water availability falls below 1,000 cubic metres.
- **Aquifer** — An underground layer of rock or sediment that holds and transmits groundwater.
- **Groundwater Recharge** — The natural or assisted process by which water percolates underground to replenish aquifers.
- **Water Footprint** — The total volume of freshwater used, directly and indirectly, by a person, product, or activity.

Complete Detailed Explanation

Part A — Why India Is Water-Stressed, In Depth

India receives roughly 80% of its annual rainfall during the four monsoon months, making storage and groundwater recharge critical — without adequate infrastructure, most of this water runs off into rivers and out to sea rather than being captured for the dry months. Rapid urbanisation has also sealed natural recharge zones under concrete, further reducing the land's ability to absorb rainwater.

Groundwater has historically absorbed the shortfall, but the Ministry of Jal Shakti's Dynamic Ground Water Resource Assessment 2024 found that annual national groundwater extraction already stands at over 60% of annual recharge, with several states and districts extracting groundwater faster than it can naturally replenish — a condition described as “over-exploited.”

Part B — The Ten Building Blocks of Module 3

This is the heart of the module. Each topic below is explained with its core meaning and its practical relevance for Indian households and communities.

1. Importance of Water Conservation

Water is essential for drinking, sanitation, agriculture, and industry, yet freshwater makes up less than 1% of all water on Earth that is readily accessible for human use. Conservation ensures adequate supply for future generations amid growing demand and climate variability.

2. Water Cycle

The water cycle — evaporation, condensation, precipitation, and collection — continuously recycles water between the atmosphere, land, and oceans. Deforestation, urbanisation, and excessive groundwater extraction can all disrupt this natural cycle.

3. Rainwater Harvesting

Rainwater harvesting collects and stores rainwater for later use through rooftop collection systems, recharge pits, or storage tanks. It reduces dependency on groundwater, recharges aquifers, and helps prevent urban flooding during heavy monsoon spells.

4. Groundwater Conservation

Groundwater is a major source of drinking and irrigation water in India, but is being depleted faster than it can naturally recharge in many regions. Conservation methods include recharge wells, check dams, afforestation, and regulated extraction for agriculture and industry.

5. Water Pollution

Water pollution occurs when harmful substances — industrial effluents, untreated sewage, agricultural runoff, and plastic waste — contaminate water bodies, making them unsafe for drinking, aquatic life, and irrigation, and a leading cause of waterborne disease.

6. Household Water Saving Practices

Simple household habits — fixing leaks promptly, turning off taps while brushing or washing, using buckets instead of hoses, and reusing greywater for plants — can cut water consumption significantly without major investment.

7. Industrial Water Management

Industries consume large volumes of water for processing and cooling. Responsible management includes treating and recycling wastewater, adopting water-efficient technologies, and complying with pollution control regulations before any discharge into water bodies.

8. River Conservation

Rivers face threats from pollution, encroachment, over-extraction, and declining flow. Conservation efforts include controlling industrial discharge, protecting floodplains, afforestation along riverbanks, and government-led river rejuvenation programmes such as Namami Gange.

9. Government Water Initiatives

National programmes such as the Jal Jeevan Mission, Atal Bhujal Yojana, and Namami Gange work to expand access to safe drinking water, encourage community-led groundwater management, and rejuvenate major rivers.

10. Daily Water Saving Checklist

Turning conservation into a daily habit — checking for leaks, reusing water where possible, and watering plants efficiently — is what ultimately determines whether national water policy translates into real savings on the ground.

Important Terminologies

Term	Simple Meaning
Water Stress	Annual per capita water availability below 1,700 cubic metres
Water Scarcity	Annual per capita water availability below 1,000 cubic metres
Aquifer	An underground layer of rock or sediment that stores and transmits groundwater
Rainwater Harvesting	Collecting and storing rainwater for later use
Groundwater Recharge	The process by which water percolates underground to replenish aquifers
Greywater	Gently used household water (from sinks, showers, laundry) that can be reused for non-drinking purposes
Water Footprint	Total freshwater used, directly and indirectly, by a person, product, or activity
Check Dam	A small barrier built across a drainage channel to slow water flow and aid recharge
Water Table	The upper level of underground water in the soil
Effluent	Liquid waste or sewage discharged into a water body, often after treatment

Step-by-Step Understanding

How Rooftop Rainwater Harvesting Works

1. Catchment — Rain falls on a rooftop or paved surface

2. Collection — Pipes or channels direct the water away from the catchment area

3. First-Flush Diversion — The initial, dirtiest flow of water is diverted away to avoid contamination

4. Filtration — Remaining water passes through a filter to remove debris

5. Storage or Recharge — Water is either stored in a tank for direct use or channelled into a recharge pit or well to replenish groundwater

How to Reduce Your Household Water Footprint

1. Fix leaking taps and pipes as soon as they appear

2. Install low-flow fixtures such as aerator taps and dual-flush toilets

3. Reuse greywater from washing for cleaning or watering plants

4. Water gardens early morning or evening to reduce evaporation loss

5. Run washing machines and dishwashers only with full loads

6. Track and review your household's monthly water usage

Real-Life Indian Examples

(Widely known public examples, described for educational awareness.)

- **Rajasthan's Bothara Village:** After implementing check dams and water-harvesting trenches under the Jal Jeevan Mission, the village saw water levels rise significantly and storage capacity increase, demonstrating the impact of simple structural interventions.
- **Mission Amrit Sarovar:** Launched in 2022, this national mission aims to develop or rejuvenate at least 75 water bodies in every district of the country as a lasting water conservation legacy.
- **Namami Gange Programme:** A major national mission to clean and rejuvenate the river Ganga through pollution control, sewage treatment infrastructure, and afforestation along the riverbank.
- **Jal Jeevan Mission Progress:** As of October 2025, more than 15.72 crore rural households — over 81% of rural India — have gained functional tap water connections, up from just 16.8% when the mission launched in 2019.

Practical Scenarios

Scenario 1: You notice a public water pipeline has been leaking on your street for several days.

Note the exact location and report it to your local municipal water department or civic helpline — a single unreported leak can waste thousands of litres over time.

Scenario 2: Your housing society is planning a new building but has not considered rainwater harvesting.

Raise it during the planning stage — many states legally require rainwater harvesting structures for new constructions above a certain plot size; check your local municipal byelaws.

Scenario 3: A nearby factory appears to be discharging coloured effluent into a local drain that feeds into a river.

Avoid contact with the water, document the location and time if safe to do so, and report it to your State Pollution Control Board.

Government Rules / Guidelines

- **Ministry of Jal Shakti:** The apex national body for water resources, drinking water supply, and sanitation, overseeing missions including Jal Jeevan Mission and Namami Gange.
- **Central Water Commission (CWC):** India's premier technical body for water resources planning, assessment, and management, publishing periodic national water availability assessments.
- **Central Ground Water Board (CGWB):** Monitors and assesses groundwater resources nationally and regulates groundwater extraction in critical areas.
- **Central and State Pollution Control Boards (CPCB / SPCBs):** Monitor water quality and enforce standards for industrial and municipal effluent discharge.

Laws & Policies

Law / Policy	What It Covers
Water (Prevention & Control of Pollution) Act, 1974	Regulates discharge of pollutants into water bodies and establishes pollution control boards
Environment (Protection) Act, 1986	The umbrella law empowering government action against any activity, including water pollution, that harms the environment
National Water Policy, 2012	Sets the national framework for water resource planning, conservation, and equitable allocation
Jal Jeevan Mission (Har Ghar Jal)	Aims to provide functional tap water connections to every rural household at 55 litres per capita per day
Atal Bhujal Yojana	A community-led scheme for sustainable groundwater management in water-stressed areas
Namami Gange Programme	A mission-mode programme to clean, conserve, and rejuvenate the river Ganga

> *Water is constitutionally a State subject in India — planning, approval, and implementation of water supply schemes rest with State and Union Territory governments, while the Centre provides technical and financial support through national missions.*

Important Statistics

Metric	Figure
India's average per capita water availability (CWC, 2024)	~1,513 cubic metres per year (below the 1,700 m³ water-stress threshold)
Projected per capita water availability by 2031	~1,367 cubic metres per year
India's share of global freshwater resources	About 4%
India's share of global population	About 18%
National groundwater extraction as a share of annual recharge (2024)	About 60.47%
Rural households with tap water connections under Jal Jeevan Mission	Over 81% (as of October 2025), up from 16.8% in 2019
Mandatory water recycling requirement for cities (Ministry of Jal Shakti)	At least 20% of consumed water to be recycled and reused

(Figures sourced from the Central Water Commission (CWC), Ministry of Jal Shakti, and Jal Jeevan Mission dashboard; official portals should be checked for the latest updates.)

Best Practices

- Fix leaking taps and pipes as soon as they are noticed
- Install or advocate for rainwater harvesting structures in your building or society
- Reuse greywater for cleaning, flushing, or watering plants where feasible
- Choose water-efficient fixtures and appliances
- Report public water leaks and pollution to the relevant authority promptly
- Support and participate in local water body rejuvenation efforts

Common Mistakes

- Leaving taps running while brushing teeth, shaving, or washing dishes
- Ignoring small leaks because they seem insignificant
- Watering gardens during the hottest part of the day, losing water to evaporation
- Assuming water conservation is only relevant during a visible drought or crisis
- Discharging untreated greywater or effluent into open drains

Myths vs Facts

Myth	Fact
"India gets enough monsoon rain, so water scarcity isn't a real concern"	Most monsoon rainfall runs off unused without adequate storage and recharge infrastructure, leaving India water-stressed despite substantial total rainfall
"Groundwater is an unlimited resource"	Groundwater is being extracted faster than it recharges in many regions, and over-extraction can permanently lower water tables
"Rainwater harvesting is only useful in dry regions"	It helps recharge groundwater and reduce urban flooding even in high-rainfall areas, and is legally required for many new buildings
"A small household leak doesn't waste much water"	A single dripping tap can waste thousands of litres of water over the course of a year

Do's and Don'ts

Do	Don't
Fix leaking taps and pipes promptly	Don't leave taps running while brushing or washing
Harvest rainwater where structurally possible	Don't ignore recurring small leaks
Reuse greywater for cleaning or plants	Don't water plants during peak afternoon heat
Report public leaks and pollution promptly	Don't discharge untreated effluent into drains
Support local water body rejuvenation efforts	Don't treat water conservation as relevant only during a crisis

Practical Tips

For Households

Fix leaks immediately • Use a mug instead of a running tap while shaving or brushing • Reuse RO reject or washing water for cleaning or plants

For Communities

Advocate for rainwater harvesting structures in housing societies • Organise or support local check-dam and pond rejuvenation projects • Report leaking public pipelines promptly

For Farmers

Consider drip or sprinkler irrigation over flood irrigation • Choose less water-intensive crops in water-stressed regions • Use farm ponds and check dams to capture monsoon runoff

Practical Checklist

■	I check my home regularly for leaking taps or pipes
■	I know whether my building has a rainwater harvesting structure
■	I reuse greywater for cleaning or watering plants where possible
■	I know which government body to contact to report a public water leak
■	I water plants in the early morning or evening rather than midday
■	I am aware of my region's groundwater stress level
■	I have discussed water-saving habits with family or peers

Quick Revision Notes

- Water stress = per capita availability below 1,700 m³/year; scarcity = below 1,000 m³/year
- India's per capita water availability (2024) is ~1,513 m³/year — already in the water-stress zone
- The water cycle: evaporation → condensation → precipitation → collection
- Groundwater extraction nationally exceeds 60% of annual recharge
- Key missions: Jal Jeevan Mission, Atal Bhujal Yojana, Namami Gange
- CWC, CGWB, and CPCB/SPCBs are India's key water-management regulatory bodies

Key Takeaways

1. India's water challenge is primarily about storage, distribution, and waste — not a lack of total rainfall

2. Groundwater is being depleted faster than it recharges in many regions, making conservation urgent rather than optional

3. Simple daily habits — fixing leaks, reusing greywater, harvesting rainwater — collectively make a measurable difference

4. Water is a State subject in India, but national missions provide critical technical and financial support

5. Reporting leaks and pollution promptly is a direct, practical form of civic participation

Glossary

- **Aquifer** — An underground layer of rock or sediment that stores and transmits groundwater
- **Check Dam** — A small barrier across a drainage channel that slows water flow and aids recharge
- **Effluent** — Liquid waste or sewage discharged into a water body
- **Greywater** — Gently used household water that can be reused for non-drinking purposes
- **Groundwater Recharge** — The process by which water percolates underground to replenish aquifers
- **Rainwater Harvesting** — Collecting and storing rainwater for later use
- **Water Footprint** — Total freshwater used, directly and indirectly, by a person, product, or activity
- **Water Scarcity** — Annual per capita water availability below 1,000 cubic metres
- **Water Stress** — Annual per capita water availability below 1,700 cubic metres
- **Water Table** — The upper level of underground water in the soil

Frequently Asked Questions

1. Why is India considered a water-stressed country?
Because its average annual per capita water availability (~1,513 cubic metres in 2024) falls below the internationally recognised water-stress threshold of 1,700 cubic metres.

2. What is the difference between water stress and water scarcity?
Water stress is availability below 1,700 cubic metres per person per year; water scarcity is a more severe condition, below 1,000 cubic metres per person per year.

3. How does rainwater harvesting help?
It captures monsoon runoff for storage or groundwater recharge, reducing dependence on over-extracted groundwater and helping prevent urban flooding.

4. Why is groundwater depletion a serious concern in India?
Because national groundwater extraction already exceeds 60% of annual recharge, with several regions extracting far more than can be naturally replenished.

5. What is the Jal Jeevan Mission?
A national mission launched in 2019 to provide functional tap water connections to every rural household in India at a minimum service level of 55 litres per capita per day.

6. What is Namami Gange?
A mission-mode national programme to clean, conserve, and rejuvenate the river Ganga through pollution control and sewage treatment infrastructure.

7. Which law regulates water pollution in India?
The Water (Prevention & Control of Pollution) Act, 1974, which established the pollution control board system.

8. Is water a state or central subject in India?
Water is constitutionally a State subject; state governments plan and implement water supply schemes, while the Centre provides technical and financial support.

9. How much water can a single dripping tap waste?
A slow but steady drip can waste thousands of litres of water over the course of a year if left unrepaired.

10. How can I report water pollution or a public leak?
Contact your local municipal water department, urban local body, or the State Pollution Control Board; many cities also offer dedicated civic grievance apps or helplines.

Official References

- Ministry of Jal Shakti — Jal Jeevan Mission, Atal Bhujal Yojana, and Namami Gange programme information
- Central Water Commission (CWC) — Assessment of Water Resources of India 2024
- Central Ground Water Board (CGWB) — Dynamic Ground Water Resource Assessment 2024
- Central Pollution Control Board (CPCB) — water quality monitoring and standards
- Water (Prevention & Control of Pollution) Act, 1974, and National Water Policy, 2012 — India Code portal

Official Government Website Links

- Jal Jeevan Mission: **https://ejalshakti.gov.in**
- Ministry of Jal Shakti: **https://jalshakti.gov.in**
- Central Water Commission: **https://cwc.gov.in**
- Central Ground Water Board: **https://cgwb.gov.in**
- Press Information Bureau: **https://pib.gov.in**

Conclusion

Water conservation is not a task for another day — it is a practice for today. India's water future depends on the collective sum of everyday choices: a fixed leak, a harvested rooftop, a reused bucket of water, a reported violation. None of these actions alone solves a national crisis, but together, at scale, they are exactly what national water security depends on.

Continue your Environment & Civic Responsibility journey with JanGyan. Stay mindful, stay consistent, and remember: every drop saved today secures tomorrow.

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